

TWO TRIANGULATIONS THAT SHARE ONLY THEIR HULL

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ABSTRACT. Any two triangulations of a finite point set in the plane contain the edges of its convex hull. We show that, for every $n \geq 9$, there are n points in general position with a pentagonal hull and two triangulations that have no other edge in common. At least five edges must be shared whenever $n \geq 5$, so this settles the minimum for all $n \geq 9$. The construction starts from an explicit set of nine points and adds points one at a time; the main step is a lemma showing that the configuration needed for the next insertion always reappears. A search over all order types of at most ten points shows that the minimum is 4, 5, 6, 6, 6, 5, 5 for $n = 4, \dots, 10$, so after the five-point case the minimum first returns to five at nine points.

1. INTRODUCTION

Let P be a set of n points in the plane, no three on a line. A *triangulation* of P is a maximal set of pairwise non-crossing segments with endpoints in P ; it divides the convex hull of P into triangles, its *faces*. No edge can cross an edge of the convex hull, so every triangulation contains every hull edge, and two triangulations of P always share at least as many edges as the hull has vertices. How much more must they share?

The question was raised on MathOverflow in 2023 [4]. Write

$$f(n) = \min_{|P|=n} \min_{A,B} |A \cap B|,$$

where P ranges over sets of n points in general position and A, B over triangulations of P , each regarded as a set of edges. Contributors to the thread observed that a triangular or quadrilateral hull with an interior point forces at least six shared edges, so that $f(n) \geq 5$ for $n \geq 5$ and five can only occur with a pentagonal hull; this also follows from the forced-edge argument of [3]. Random searches found pairs sharing six edges for $8 \leq n \leq 11$, but none sharing five.

Theorem 1. *For every $n \geq 9$ there is a set of n points in general position with a pentagonal hull and two triangulations whose only common edges are the five hull edges. Hence $f(n) = 5$ for all $n \geq 9$.*

The question is related to geometric thickness. A graph is *doubly linear* if it can be drawn with straight edges on some point set so that its edges split into two non-crossing classes; Hutchinson, Shermer, and Vince [3] showed that such a graph on $n \geq 4$ vertices has at most $6n - 18$ edges. For a point set with a triangular hull, two triangulations sharing s edges have a union of $6n - 12 - s$ edges, so the bound says $s \geq 6$ there, and equality would require a doubly linear graph with $6n - 18$ edges. Durocher, Gethner, and Mondal [2] constructed doubly linear graphs with $6n - 19$ edges for every $n \geq 9$; completing their two layers to triangulations shows $f(n) \leq 7$.

Section 2 proves the lower bound. Section 3 gives the nine-point example, and Sections 4 and 5 show how to extend it. Section 6 explains why the answer changes at nine.

2. THE LOWER BOUND

A segment between two points of P is *unavoidable* if every triangulation of P contains it. Hull edges are unavoidable, and so is any segment that no other segment crosses. The lemma and corollary below are a version of the forced edges of Hutchinson, Shermer, and Vince [3, Lemma 5 and the proof of Theorem 6]; we include the short proofs to keep the paper self-contained.

Lemma 2. *Let u be a hull vertex of P and m a line through u that has all other points of P strictly on one side. If q is a point of $P \setminus \{u\}$ nearest to m , then uq is unavoidable.*

Proof. Let ℓ be the line through q parallel to m , and H the closed half-plane bounded by ℓ that does not contain u . All points of P other than u lie in H . Every point of the open segment uq lies strictly between m and ℓ , hence outside H . A segment crossing uq does so at such a point, so it is not contained in H ; but a segment with both endpoints in H lies in H , and a segment with an endpoint at u does not cross uq . So no segment crosses uq . $\square$

Corollary 3. *If $|P| \geq 5$, two triangulations of P share at least five edges, and if they share exactly five then the hull of P is a pentagon.*

Proof. If the hull has at least five vertices, its edges suffice. Otherwise P has an interior point. If the hull is a triangle uvw , apply Lemma 2 at u with m parallel to vw ; interior points are closer to m than v and w are, so the resulting segment is interior. Doing the same at v and at w gives three interior unavoidable segments, which are distinct because their hull endpoints differ. With the hull edges that makes six. If the hull is a quadrilateral uvw , one of the triangles uvx and wvx contains an interior point; if it is uvx , apply the lemma at u with m parallel to vx , and otherwise at w . This gives an interior unavoidable segment ending at u or w . The diagonal uw gives, in the same way, one ending at v or x . With the four hull edges that makes six. $\square$

3. NINE POINTS

Number the points as follows.

i	1	2	3	4	5	6	7	8	9
x_i	33	47	47	29	28	21	23	2	0
y_i	47	30	5	20	26	21	29	0	31

Write ij for the segment from point i to point j . The hull is the pentagon 1, 2, 3, 8, 9, with edges 12, 23, 38, 89, 91. Figure 1 shows the triangulations

$$\begin{aligned} A &= \text{hull} \cup \{14, 15, 16, 17, 18, 24, 28, 45, 48, 56, 58, 67, 68, 78\}, \\ B &= \text{hull} \cup \{25, 27, 29, 34, 35, 36, 37, 39, 46, 47, 49, 57, 69, 79\}. \end{aligned}$$

The two sets of interior edges are disjoint. One checks from the coordinates that no three points are collinear and that no two edges of A , or of B , cross. Each of A and B then has 19 pairwise non-crossing edges, and $19 = 3 \cdot 9 - 3 - 5$ is the number of edges of any triangulation of nine points with five on the hull (Euler's formula), so both are triangulations. Hence $f(9) \leq 5$.

4. ADDING A POINT

Call two triangulations of the same point set *hull-disjoint* if they share only the hull edges.

Definition 4. Let A and B be triangulations of P . A face α of A and a face β of B form a *good pair* if their interiors intersect and they have no vertex in common.

For example, in Figure 1 the face 156 of A and the face 479 of B form a good pair; the point (24, 24) lies inside both.

Lemma 5 (Insertion). *Let A and B be hull-disjoint triangulations of P with a good pair (α, β) , and let $p \in \text{int}(\alpha) \cap \text{int}(\beta)$ lie on no line through two points of P . Let A' be A with p joined to the three vertices of α , and B' be B with p joined to the three vertices of β . Then A' and B' are hull-disjoint triangulations of $P \cup \{p\}$.*

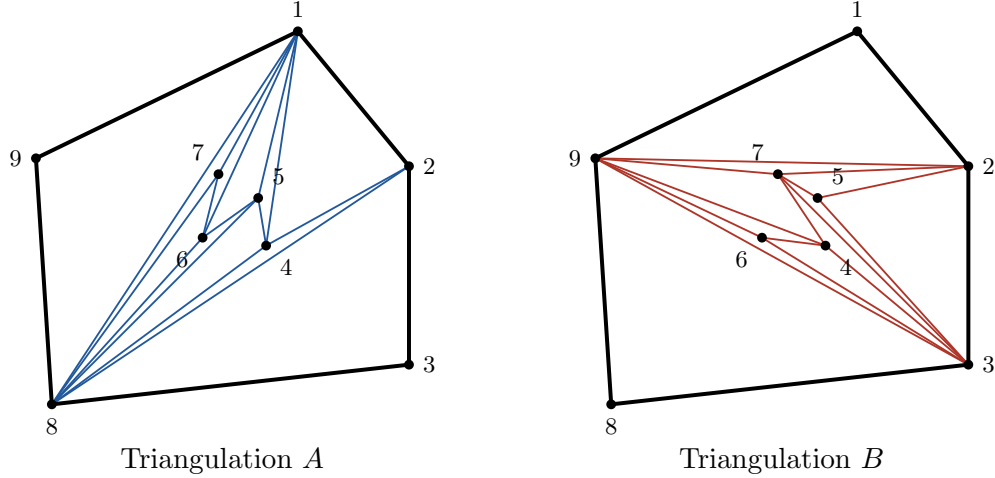


Figure 1. Two triangulations of nine points whose only common edges are the five hull edges (black).

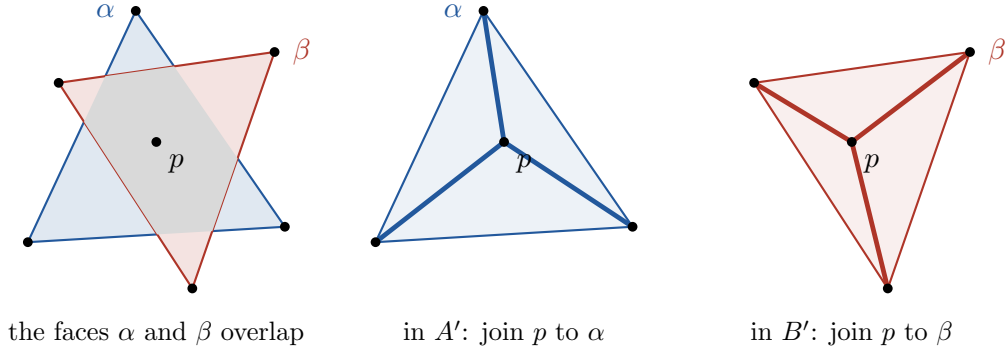


Figure 2. Lemma 5. The point p lies in the overlap of α and β (shaded). The three thick blue edges are added to A and the three thick red edges to B .

Proof. Joining a point inside a face to its three vertices cuts that face into three triangles and leaves everything else unchanged, so A' and B' are triangulations; the hull is the same as before. The old edges of A' and B' are shared only on the hull. The new edges of A' join p to the vertices of α , and those of B' join p to the vertices of β ; as α and β have no common vertex, no new edge is shared. A suitable p exists because $\text{int}(\alpha) \cap \text{int}(\beta)$ is a nonempty open set and only finitely many lines are excluded. $\square$

5. THE GOOD PAIR RETURNS

A single insertion is easy; to repeat it, a new good pair is needed. We first record two properties of a good pair (α, β) .

- (a) *No vertex of β lies in α , and no vertex of α lies in β .* A face contains no point of P other than its vertices, and by general position no other point of P lies on a side of a face.
- (b) *At least two sides of α meet $\text{int}(\beta)$, and at least two sides of β meet $\text{int}(\alpha)$.* By (a), each vertex of the convex polygon $\alpha \cap \beta$ is a crossing of a side of α with a side of β , and near that vertex the polygon is the angle between those two sides. Its sides therefore alternate between sides of α and sides of β , so it has at least four. Different sides of a convex polygon

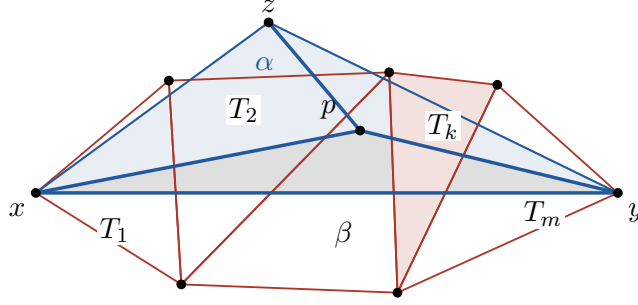


Figure 3. The case $m \geq 4$. The side xy of $\alpha = xyz$ passes through the faces $T_1, T_2, \beta, T_k, T_m$ of B (red; faces of B away from xy are omitted, and β is drawn before it is subdivided). After p is inserted, the face pxy of A' (grey) and the face T_k (shaded) form a good pair.

lie on different lines, so two different sides of α contribute sides to $\alpha \cap \beta$, and the relative interior of each such contribution lies in $\text{int}(\beta)$.

Lemma 6 (Regeneration). *Let $|P| \geq 7$, let A and B be hull-disjoint triangulations of P with a good pair (α, β) , and let A', B' be obtained as in Lemma 5. Then A' and B' have a good pair.*

Proof. Let $\alpha = xyz$ and let xy be a side of α meeting $\text{int}(\beta)$. The open segment xy passes through no point of P and is not an edge of B (it enters a face of B). Let $T_1, \dots, T_m$ be the faces of B that it meets, in order from x to y . A face of B with vertex x that meets xy contains an initial part of xy , since faces are convex; so T_1 is the only one with vertex x , and likewise T_m is the only one with vertex y . By (a), β has neither vertex, so $\beta = T_\ell$ with $1 < \ell < m$.

Case $m \geq 4$. Pick k with $1 < k < m$ and $k \neq \ell$. Then T_k is a face of B' , since only β was subdivided. Near a point of xy inside T_k , the face pxy of A' contains a half-disc, so $\text{int}(pxy) \cap \text{int}(T_k) \neq \emptyset$. The vertices of T_k are points of P other than x and y , and $p \notin P$. So (pxy, T_k) is a good pair (Figure 3). The same argument applies with the roles of A and B exchanged.

Remaining case. Suppose that every side of α meeting $\text{int}(\beta)$ meets exactly three faces of B , and every side of β meeting $\text{int}(\alpha)$ meets exactly three faces of A . We show that $|P| = 6$.

Step 1: each side of α through β has an ear behind it. Let xy meet $\text{int}(\beta)$. It enters β across a side g_1 and leaves across a side $g_2 \neq g_1$, so its three faces are T_1 , the triangle on g_1 with third vertex x ; then β ; then T_3 , the triangle on g_2 with third vertex y . Since xy is not a hull edge, there is a face $D = xyd$ of A on the other side of xy from α . The side g_1 crosses xy , so it meets $\text{int}(\alpha)$ and passes from α into D . By assumption g_1 meets exactly three faces of A : α in the middle, and at each end the triangle formed by an endpoint of g_1 and the side of α that g_1 crosses there. One of these is D , so d is an endpoint of g_1 . The same argument with g_2 shows that d is an endpoint of g_2 . Hence $d = b$, the common vertex of g_1 and g_2 . Now xb is an edge of D and of T_1 , and yb is an edge of D and of T_3 . Shared edges are hull edges, so b is a hull vertex whose neighbours on the hull are x and y (Figure 4).

Step 2: two ears. By (b), two sides of α meet $\text{int}(\beta)$; call them xy and yz . Step 1 gives hull vertices b , with hull neighbours x and y , and b' , with hull neighbours y and z . They differ, since a hull vertex has only two hull neighbours, and both are vertices of β , so $\beta = bb'c$ for some c . The two sides of β that xy crosses are the two sides at b , namely bb' and bc .

Step 3: the hexagon. The side bc crosses xy , entering α , and leaves α across yz or zx , because c lies outside α . By assumption the face of A beyond the exit side has c as its third vertex. If the exit side is yz , that face is yzb' by Step 1, so $c = b'$, which is impossible. If the exit side is zx , then zx meets $\text{int}(\beta)$, and Step 1 for zx shows that c is a hull vertex with hull neighbours z and x . Then x, b, y, b', z, c are consecutive hull vertices forming a closed cycle, so the hull is the hexagon

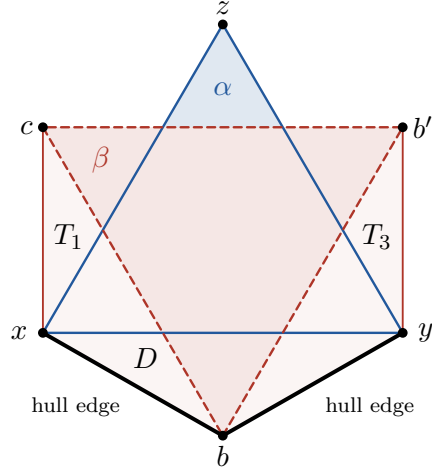


Figure 4. Step 1. The side xy of α passes through T_1 , β (dashed) and T_3 . The face D of A beyond xy has third vertex b , so xb and yb lie in both triangulations and must be hull edges (thick).

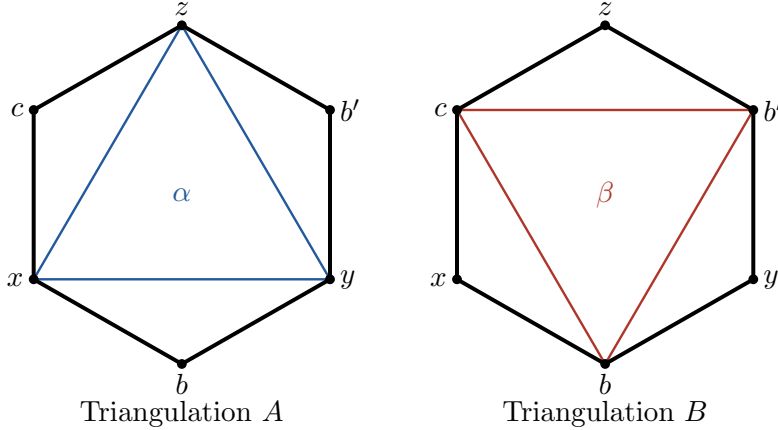


Figure 5. The only configuration in which a good pair can fail to return: six points, with A made of $\alpha = xyz$ and three ears and B made of $\beta = bb'c$ and three ears. After a point is inserted in $\alpha \cap \beta$, no good pair remains.

$xb y b' z c$, and the faces α , xyb , yzb' , zxc of A cover it. These faces contain no further points, so $|P| = 6$ (Figure 5). $\square$

Proof of Theorem 1. The triangulations of Section 3 are hull-disjoint and have a good pair. By Lemmas 5 and 6, points can be added one at a time, keeping the pentagonal hull, hull-disjoint triangulations, and a good pair. This gives $f(n) \leq 5$ for every $n \geq 9$, and Corollary 3 gives $f(n) \geq 5$. $\square$

At each step the point p can be chosen rational, so for each n the construction yields integer coordinates after clearing denominators.

6. WHY NINE

Aichholzer, Aurenhammer, and Krasser [1] listed all order types of up to ten points in general position. Two point sets have the same order type if some bijection between them preserves the

orientation of every triple or reverses all of them, so a set and its mirror image count once; they then have the same crossing pairs of segments and hence the same triangulations. There are 3,315 order types of eight points, 158,817 of nine and 14,309,547 of ten.

For a single point set the minimum is a finite computation. Two triangulations of P have $2(3n - 3 - h)$ edges counted with multiplicity, where h is the number of hull vertices, so they share $2(3n - 3 - h) - |A \cup B|$ edges; and $A \cup B$ is a set of segments that splits into two non-crossing classes. Conversely, each class of such a split extends to a triangulation. So the minimum number of shared edges is

$$6n - 6 - 2h - \binom{n}{2} + \text{oct}(P),$$

where $\text{oct}(P)$ is the least number of segments that must be removed from the crossing graph of all $\binom{n}{2}$ segments to make it bipartite. Evaluating this for every order type gives, for point sets in general position,

n	4	5	6	7	8	9	10
$\min A \cap B $	4	5	6	6	6	5	5

So the minimum rises to six and returns to five at $n = 9$. Exactly seven order types of nine points and 734 of ten attain five, all with pentagonal hulls; the set of Section 3 is one of the seven. None of the seven has a nontrivial symmetry: for each, the only bijection to itself that preserves or reverses every orientation is the identity. So no nine-point example is symmetric. For a triangular hull the minimum is seven when $n = 9$ and when $n = 10$, which means that the largest doubly linear graphs on nine or ten points in general position have $6n - 19$ edges.

We end with two questions. Is the minimum for a triangular hull equal to seven for every $n \geq 9$? And can a set of six, seven, or eight points with collinear triples, which the enumeration does not include, have two triangulations sharing only five edges?

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